

Termeh Taheri

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Profile

ML / Research Engineer specializing in audio + music ML, GPU-accelerated PyTorch experimentation, and scalable model pipelines. Experience with distributed training (DDP/FSDP), profiling, and research prototyping. Built production-grade FastAPI + Docker + AWS systems and integrated text-to-music models into reproducible research workflows.

Experience

AI Engineer, London College of Business Studies

Jul 2025 - Present

- Built and deployed production-grade FastAPI + Docker services on AWS EC2 to orchestrate multimodal content-generation workflows around generative models.
- Designed and optimized inference pipelines (batching, caching, async I/O), improving latency and throughput for large-scale LLM-driven generation pipelines.

Software Engineer, Roshan Company

Aug 2022 - Aug 2024

- Built and maintained high-performance Python services (Django, FastAPI, Docker) supporting ML-driven products and large-scale data workflows.
- Optimized data pipelines and CI/CD deployments (profiling, batching, caching, indexing), reducing system latency and enabling reproducible releases.

Selected Projects

CMI-Arena – Large-Scale Evaluation Platform for Text-to-Music Models

- Integrated and containerized text-to-music models (Lyria RealTime, DiffRhythm, etc.) as Dockerized FastAPI services for reliable GPU-backed inference.
- Built streaming audio generation pipelines with optimized I/O, batching, and request handling for multi-model evaluation at scale.

SAR-LM – Symbolic Audio Reasoning with Large Language Models ([Code](#))

- Built a modular PyTorch pipeline for audio feature extraction (PANNs, Whisper, MT3, Chordino) and reproducible benchmarking on MMAU/MMAR/OmniBench, enabling large-scale audio reasoning experiments and interpretable model analysis.

Publications

SAR-LM: Symbolic Audio Reasoning with LLMs – Accepted at LLM4MA Workshop @ ISMIR 2025; under review at ICASSP 2026. ([Paper](#))

OmniVideoBench: Towards Audio-Visual Understanding Evaluation for Omni MLLMs – Under review at ICLR 2026. ([Paper](#))

Breast Cancer Prediction by Ensemble Meta-Feature Space Generator Based on Deep Neural Network – Biomedical Signal Processing and Control (2024). ([Paper](#))

Presentation of Encryption Method for RGB Images Based on an Evolutionary Algorithm Using Chaos Functions and Hash Tables – Multimedia Tools and Applications (2023). ([Paper](#))

Education

MSc Artificial Intelligence – Queen Mary University of London (Distinction)

September 2025

Thesis: SAR-LM: Symbolic Audio Reasoning with Large Language Models

Advisor: Prof. Emmanouil Benetos

BSc Computer Engineering – Noshirvani University of Technology

July 2022

Thesis: Ensemble Deep Learning Algorithm for Medical Image Analysis

Software Proficiency

ML & Deep Learning: Python, PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn, Keras | **Audio & Signal Processing:** Librosa, Essentia, FFmpeg, OpenCV | **Backend & Systems:** FastAPI, Django, Docker, Linux, Celery, Redis, PostgreSQL, Elasticsearch, Pytest | **Cloud & Deployment:** AWS (EC2), CI/CD | **Other:** Java, C++, MATLAB, React, HTML, CSS

Honors & Awards

Chevening Scholarship – Full scholarship recipient for MSc Artificial Intelligence, QMUL

2024–2025

Best Bachelor Project Award, Noshirvani University of Technology – First place across all departments

2022

Research Grant (No P/M/1110) – Supported publication in *Biomedical Signal Processing and Control*

2021